



Deependra Kumar



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EDUCATION & SCHOLASTIC ACHIEVEMENTS

Program	Institution	CGPA / %	YoC
MTech in Computational & Experimental Mechanics	Indian Institute of Technology Madras, Tamil Nadu	8.58	2024
BTech in Chemical Engineering	Indian Institute of Technology Gandhinagar, Gujarat	7.18	2021
Intermediate (Class XII)	National Inter College Maudaha Hamirpur, UP	88.00 %	2016
Highschool (Class X)	Aadarsh Inter College Maudaha Hamirpur, UP	83.34 %	2014
- One of the shortlisted candidates for Executive Trainee 2024 in the Chemical stream, among 876 candidates at NPCIL - Scored 93.40 percentile score, ranking in the top 7% among 19,345 applicants in the Graduate Aptitude Test in Engineering (GATE) 2022 - Received Gold Medal for Outstanding Performance in Athletics at the 10th Convocation of IIT Gandhinagar Awarded the 'Scholarship for Excellence in Sports and Games' for the academic year 2020-21 - Qualified in both JEE Main with 94.16 percentile and Advanced exams in 2017, securing admission to IIT Gandhinagar			

PROFESSIONAL WORK EXPERIENCE

1. Data Analyst (Electra Wheeler) <i>Python, EDA, Excel, Power BI, SQL, Market research</i>	- Analyzed and cleaned 1000+ electric 2-wheeler sales records to uncover trends and insights by region, state, and manufacturer, and develop an interactive dashboard 07/2024 - 10/2024 - Reviewed and screened potential companies for strategic partnerships to enhance the company's platform
2. Data Analyst (InnoByte) <i>Python, EDA, Excel, PostgreSQL, Power BI</i>	- Optimized and pre-processed datasets exceeding 100,000 records to enhance accuracy and reliability; conducted EDA to identify key trends and insights 08/2024 - Built dashboards with real-time KPIs, driving data-driven decisions and enhancing business performance
3. Data Entry Analyst (Toddle) <i>Excel, Python, Word</i>	Extracted data from DOCs, PDFs, and Excel files using Excel and Python, seamlessly integrating it into the company platform with 100% accuracy to enhance operational efficiency 06/2022 - 08/2022
4. Project Assistant (IITGN) <i>MS-office, Research</i>	Collaborated with JRF, SRF, and PhD students on industrial wastewater research projects in partnership with DSIR and CRDTH-affiliated 25+ industries and institutes 07/2021 - 10/2021

MTECH THESIS

1. Droplet Impact Dynamics on Porous Surfaces - Final (MATLAB, Image Processing, High-Speed Imaging)	05/2023 - 05/2024
- Leveraged a high-resolution digital and high-speed Phantom V1212 camera for capturing precise droplet impact on porous paper surfaces, and in-depth analysis of droplet impact dynamics for Weber numbers ranging from 100 to 490 - Presented "<u>Cost-effective High-speed Imaging of Droplet Impact</u>" at the 10th International and 50th National Conference on FMFP, IIT Jodhpur, Dec 2023 among 700+ exhibitors and co-authored a conference paper for the ICEM 2024 conference	

PROJECTS

1. Flight Dashboard and Analytics (Python, Streamlit, MySQL)	10/2024
- Engineered a web application to analyze and visualize flight data from 10,683 records, enabling users to search by departure and arrival cities - Developed visual analytics for flight price distribution, airline frequency, average duration, peak departure times, and daily price trends	
2. Image Classification Model for CIFAR Dataset (Python & its ML and DA libraries, CNN, ANN TensorFlow/Keras)	08/2024
Modelled a CNN for CIFAR-10 classification of 60k images, achieving 93.57% accuracy after 10 epochs, and compared it with an ANN model	
3. Heart Failure Risk Prediction (Python & its ML and DA libraries)	08/2024
Designed and implemented 5 machine learning classification models to predict heart disease using the dataset of 918 records, applying data exploration, feature engineering, and model selection techniques, achieving 83.37% accuracy with SVM	
4. Data Analysis and Visualization of Amazon Sales (Python & its ML and DA libraries, Power BI, PostgreSQL, Excel)	08/2024
- Investigated 129,000+ sales records to reveal trends, leading shipping cities/states, categories, size and other key metrics - Created a Power BI dashboard to visualize sales trends, top products, and customer preferences, and validated data accuracy through SQL	
5. A Comparative Analysis of Job Application Data (Python, NumPy, pandas, matplotlib, seaborn, Excel)	03/2024
Processed and transformed datasets of 500+ records, performed EDA to uncover trends, and validated findings through replication, demonstrating proficiency in diverse data analysis techniques	
6. Mathematical Foundations of Data Science (MFDS) Course Project (MATLAB, Statistics)	04/2023 - 05/2023
Created a live MATLAB interactive report of data science questions for a group project in the MFDS course. The report included 6 questions from 3 different categories, each with 5 to 10 variants generated using MATLAB	

PROFESSIONAL SKILLS

Database Skills:	Languages & Tools:	ML & DA Libraries:	Web Frameworks:	Others:
PostgreSQL/MySQL, MongoDB, GSheet/Excel, Power BI	Python, C++, JS, MATLAB, VS Code, Anaconda	NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, TensorFlow, Keras	HTML, CSS, Streamlit	Jupyter Notebook, Latex, MS-Office

POSITION OF RESPONSIBILITY

Class Representative: Computational & Experimental Mechanics, IIT Madras	Athletics Convenor: Acting Athletics Convener, IIT Madras
Teaching Assistant: Foundations of Fluid Mechanics (AM2530), IIT Madras	Athletics Secretary: IIT Gandhinagar Sports

CERTIFICATIONS

1. Machine Learning Specialization Coursera: Andrew Ng, Stanford Supervised Learning, Neural Networks, Unsupervised Learning, Recommender Systems, Reinforcement Learning	3. SQL Master Class: SQL for Data Analytics Udemy Advanced SQL Queries for Data Analytics, including Complex Joins, Subqueries, window functions, and Performance Optimization
2. Data Science & Machine Learning Course Udemy Python programming, data science libraries, supervised learning, unsupervised learning, and ensemble learning algorithms	4. Data Structures and Algorithms Course in Python Udemy Adept at leveraging linear and non-linear data structures for effective data handling and algorithm design

EXTRA-CURRICULAR SPHERE

- Represented IIT Madras in 55th and 56th Inter-IIT Sports Meets, winning gold in 400m hurdles (56th) and bronze (55th), while placing 4th in 400m (56th), and 4x400m relay in both Sports Meets Gold in 400m hurdles & Bronze in 400m in Sportsfest'23 and Silver Medal in 400m in Sportsfest'22 organized by Institute Gymkhana - Earned the title of the best athlete in Schroeter'23 by securing 4 gold and 1 silver medal in athletics, an event organized by Institute Gymkhana Gold in Powerlifting and Silver Medal in Road Race in Dean-Trophy'23 (Non-Inter IIT Inter Hostel Sports Fest) - Represented IIT Gandhinagar in 53rd & 54th Inter-IIT Sports Meet, and in 2nd & 3rd Inter-IIT Cultural Meet, achieved 5th place in 400m (54th)	
YoC: Year of Completion Sportsfest: An Inter-Collegiate Invitation Tournament Inter-IIT Sports Meet: National Level sports competition among IITs	